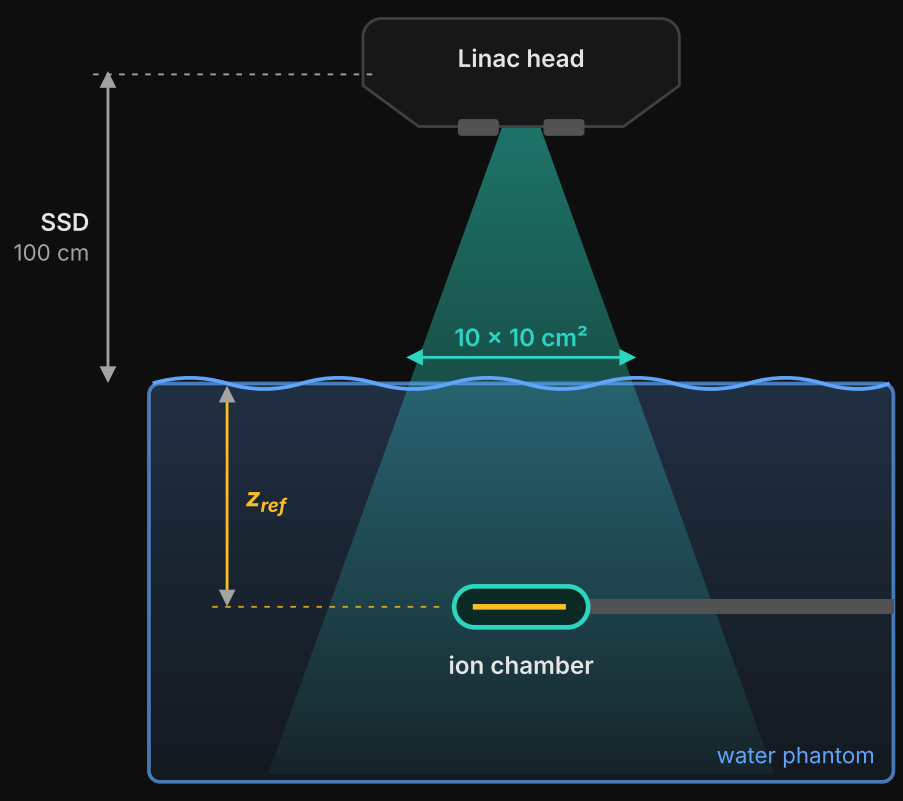
IAEA TRS-398 · CONCEPTS IN PICTURES

What each factor really means

The physics behind the dose equation, one picture at a time.

THE SETUP

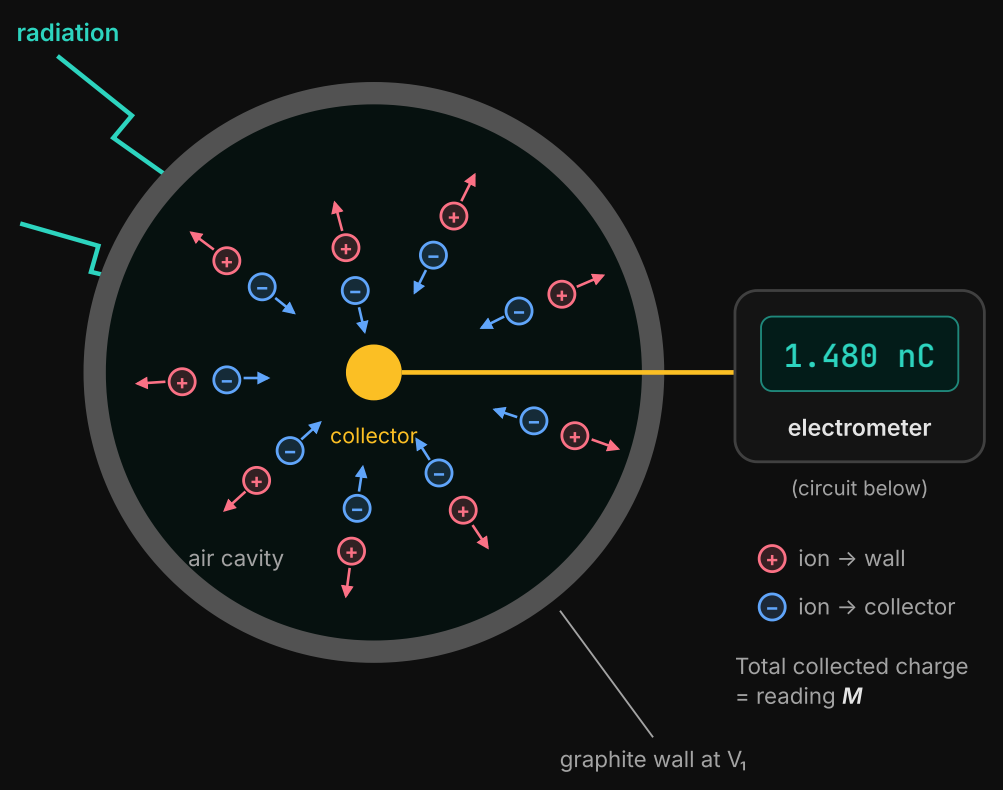
Chamber in water at the reference depth



The beam enters a water tank at a fixed distance (SSD or SAD 100 cm) with a 10×10 cm² field. The chamber sits at z_{ref}: 10 g/cm² for photons, $0.6 \cdot R_{50} - 0.1$ for electrons.

THE MEASUREMENT

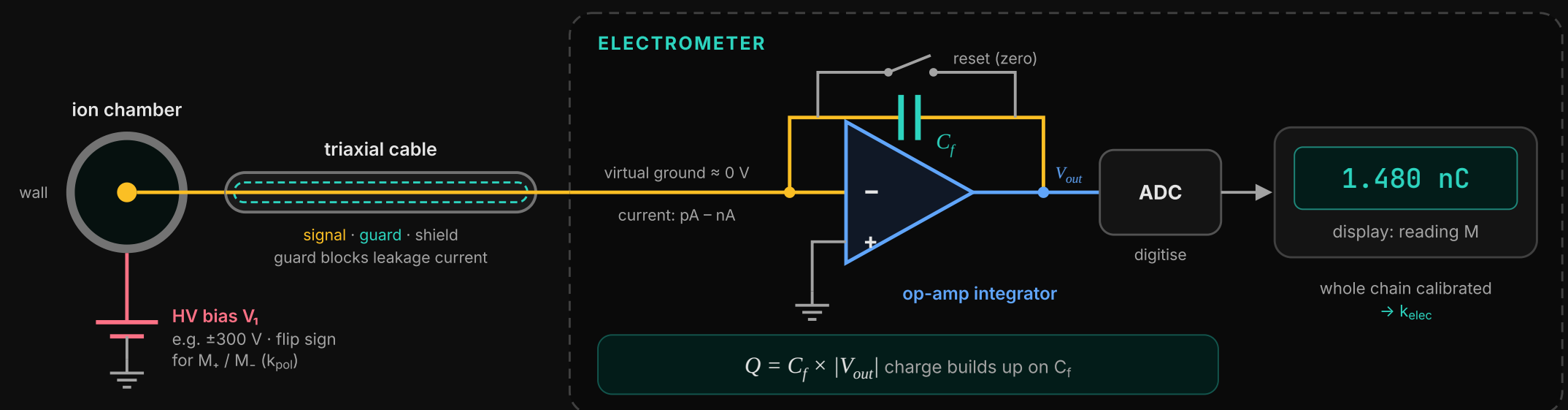
Inside the ionization chamber



Radiation knocks electrons out of the air in the cavity, creating **ion pairs**. The voltage pulls them apart: one sign to the wall, the other to the collector. The electrometer adds up that charge as the reading **M**.

THE ELECTROMETER

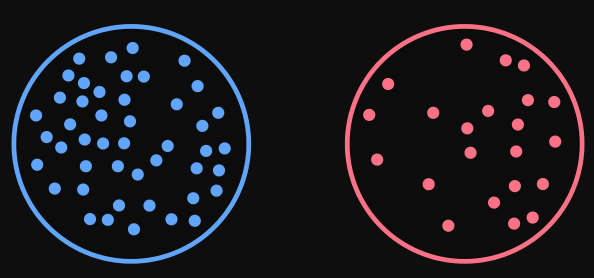
How the tiny chamber current becomes a reading



The high voltage on the chamber wall sweeps the ions out. The collector current (picoamps to nanoamps) flows down the signal wire of a **triaxial cable** into an **op-amp integrator**, whose input sits at virtual ground. The charge piles up on the feedback capacitor C_f, so the output voltage is proportional to the collected charge. The ADC digitises it and the display shows **M** in nC. The reset switch empties C_f before each measurement.

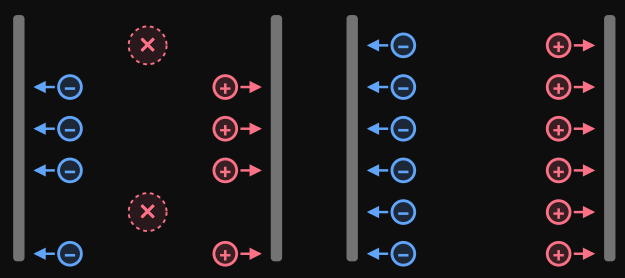
AIR DENSITY

k_{TP} temperature & pressure



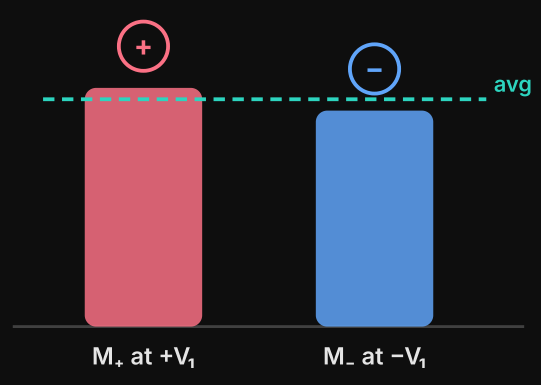
ION LOSS

k_s recombination



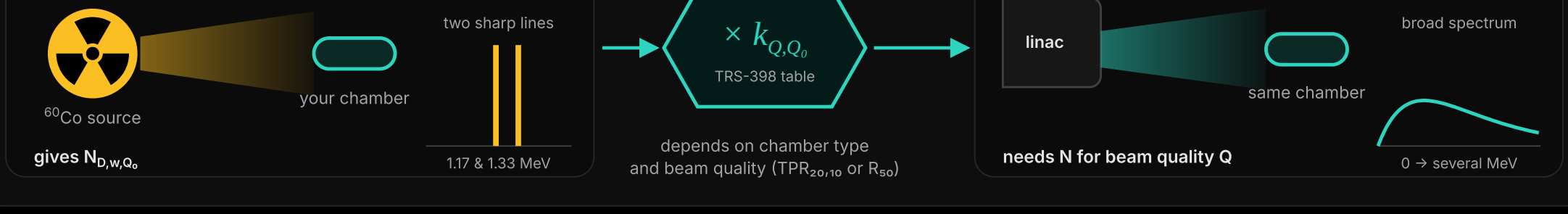
POLARITY

k_{pol} polarity effect



BEAM QUALITY

k_{Q,Q_0} translating the calibration to your beam



PUTTING IT TOGETHER

From radiation to dose

